



Q Type brand or model here, Find more manuals

SEARCH



ManualsPro / ebyte / EBYTE E22-900M33S SX1262 868/915 MHz 2Watts SPI SMD LoRa
Module User Manual

EBYTE E22-900M33S SX1262 868/915 MHz 2Watts SPI SMD LoRa Module User Manual

🕒 June 1, 2024 📁 ebyte

Table of Contents

1. [EBYTE E22-900M33S SX1262 868/915 MHz 2Watts SPI SMD LoRa Module](#)
2. [Product Information](#)
3. [Disclaimer And Copyright Notice](#)
4. [Overview](#)
5. [Parameters](#)
6. [Mechanical Dimensions and Pin Definitions](#)
7. [Notes For Hardware And Software](#)
8. [MCU Connection](#)
9. [Frequently Asked Questions](#)
10. [Welding Operation Guidance](#)
11. [Related Models](#)
12. [Antenna Guide](#)
13. [Standard Package](#)
14. [Contact Us](#)
15. [References](#)
16. [Read User Manual Online \(PDF format\)](#)
17. [Download This Manual \(PDF format\)](#)

EBYTE E22-900M33S SX1262 868/915 MHz 2Watts SPI SMD LoRa Module



Product Information

Specifications

- **Product Name:**
 - E22-900M33S
- **Description:**
 - SX1262 868/915 MHz 2Watts SPI SMD LoRa Module

- **Manufacturer:**

- Chengdu Ebyte Electronic Technology Co., Ltd.

Frequently Asked Questions

- **Q:** Can I use the E22-900M33S module without an MCU driver?
 - **A:** No, since it is a pure RF transceiver module, it requires an MCU driver or a dedicated SPI debugging tool for operation.

Disclaimer And Copyright Notice

The information in this article, including URL addresses for reference, is subject to change without notice. Documentation is provided “as is” without warranty of any kind, including any warranty of merchantability, fitness for a particular purpose, or non-infringement, and any warranty mentioned elsewhere in any proposal, specification or sample. This document disclaims all liability, including liability for infringement of any patent rights arising from the use of the information contained in this document. No license, express or implied, to the use of any intellectual property rights is granted herein by estoppel or otherwise. The test data obtained in this article are all obtained from Ebyte laboratory testing, and the actual results may be slightly different.

All trade names, trademarks and registered trademarks mentioned in this article are the property of their respective owners and are hereby acknowledged.

The final right of interpretation belongs to Chengdu Yibaite Electronic Technology Co., Ltd.

Notice

The contents of this manual may change due to product version upgrades or other reasons. Ebyte Electronic Technology Co., Ltd. reserves the right to modify the contents of this manual without any notice or prompt. This manual is only used as a guide. Chengdu Yibaite Electronic Technology Co., Ltd. makes every effort to provide accurate information in this manual. However, Chengdu Yibaite Electronic Technology Co., Ltd. does not ensure that the content of the manual is completely error-free. All statements in this manual , information and recommendations do not constitute any express or implied warranty.

Overview

Introduction

E22-900M33S is independently developed based on the new generation LoRa™ radio frequency chip SX1262 produced by Semtech in the United States . It has a maximum power of 2 W and is suitable for 868 / 915MHz patch-type LoRa™ wireless modules . It uses an industrial-grade high-precision 32MHz crystal oscillator.

Since the imported SX1262 is used as the core of the module, a power amplifier (PA) and a low noise amplifier (LNA) are built-in on the original basis , so that the maximum transmit power reaches 2 W and the receiving sensitivity is further improved. In the overall communication The stability is greatly improved compared to products without power amplifier and low noise amplifier. Compared with the previous generation LoRa™ transceiver, the anti- interference performance and communication distance have been improved, further widening the gap with products with FSK and GFSK modulation methods . This product can cover an ultra-wide applicable frequency range of 850 to 930 MHz and is backward compatible with SX1278 and SX1276. Since this module is a pure radio frequency transceiver module, it needs to use an MCU driver or a dedicated SPI debugging tool.

Features

- S X1278 module, the SX1262 module has the significant advantages of lower power consumption, faster speed, and longer distance;
- Under ideal conditions, the communication distance can reach 1 6 km;
- Built-in PA +LNA , greatly improving communication distance and communication stability;
- The maximum transmit power is 2 W, multi-level software adjustable;
- Support global license-free ISM 868/915 MHz frequency band;
- Supports multiple modulation modes, LoRa™ /GFSK ;
- LoRa™ mode supports data transmission rates from 0.3 k to 62.5 kbps ;
- G FSK mode supports data transmission rate up to 300kbps ;
- Backwards compatible with SX1278/SX1276 series RF transceivers;
- FIFO has large capacity and supports 256Byte data cache ;
- SF5 spreading factor introduced to support dense networks ;
- Supports 3.3 ~ 5.5V power supply, and any power supply greater than 5 V can ensure the best performance;
- Industrial grade standard design, supports long-term use at -40 ~ + 85 °C;
- Optional dual antennas (IPEX/stamp hole) facilitate user secondary development and integration;

Application scenarios

- Home security alarm and remote keyless entry;
- Smart home and industrial sensors, etc.;
- Wireless alarm security system ;
- Building automation solutions;
- Wireless industrial grade remote control;
- Advanced Meter Infrastructure (AMI);
- Automotive industry applications.

Parameters

Limit parameters

The main parameters

| performance|

Remark

---|---|---

minimum value| maximum value

Supply voltage (V)| 3.3| 5.5| Exceeding 5.5V will permanently burn the module

Blocking power (dBm)| –| 10| Less likely to be burned if used at close range

Working temperature (°C)| -40| +85 _| Industrial grade

Working parameters

The main parameters	performance	Remark
minimum value	Typical value	maxim um value
Working voltage (V)	3.3	5.0
Communication level (V)		3.3
Working temperature (°C)	-40	–
Working frequency band (M Hz)	850	868/915

Power consumption

| Emission current

(mA)

| | 1200| | ±100mA(33dBm&5V)

Receive current

(mA)

| | 1 5| |

Sleep current (u

A)

| | 2| | Software shutdown

Maximum transmit power (dBm)

|

32.5

|

33.0

|

33.5

| The output power of the RF chip cannot be greater than 9dBm

Receiving sensitivity (dBm)| -1 34| -1 35| -1 36| LoRaSF9

The main parameters	describe	Remark
reference distance	16 km _	Sunny and open space, antenna gain 5dBi, antenna

height 2.5 meters, air rate 0.3kbps

FIFO| 256Bytes| Maximum length of a single send

Crystal frequency| 32MHz|

Modulation| LoRa

(recommended)

|

—

Packaging method| SMD type|

Interface mode| 2.54mm| stamp hole

Communication Interface| SPI| 0 ~ 10Mbps

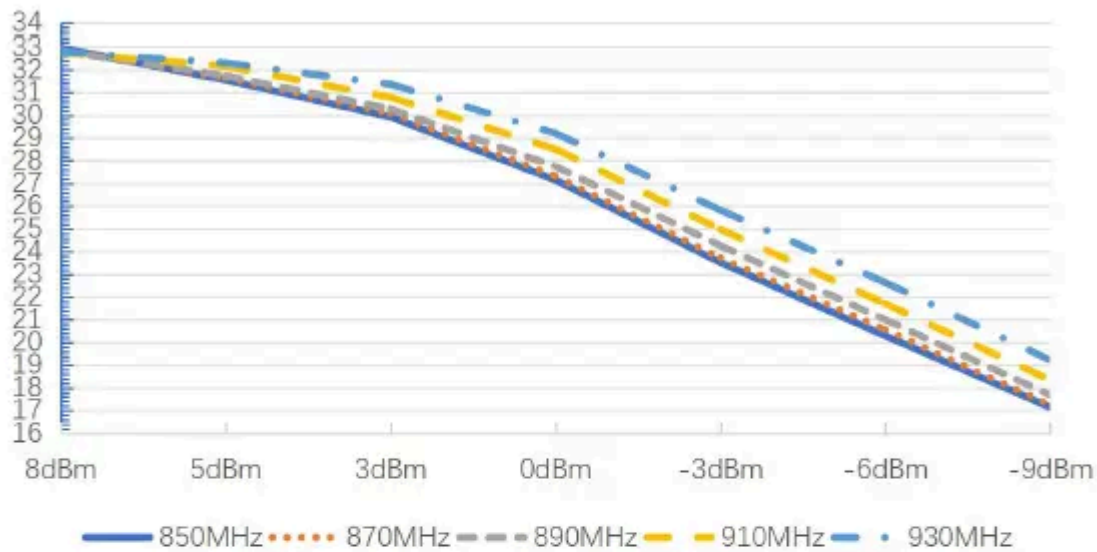
Dimensions| 38.5*24mm|

Antenna interface| Stamp hole/IPEX| Equivalent impedance is about 50 Ω

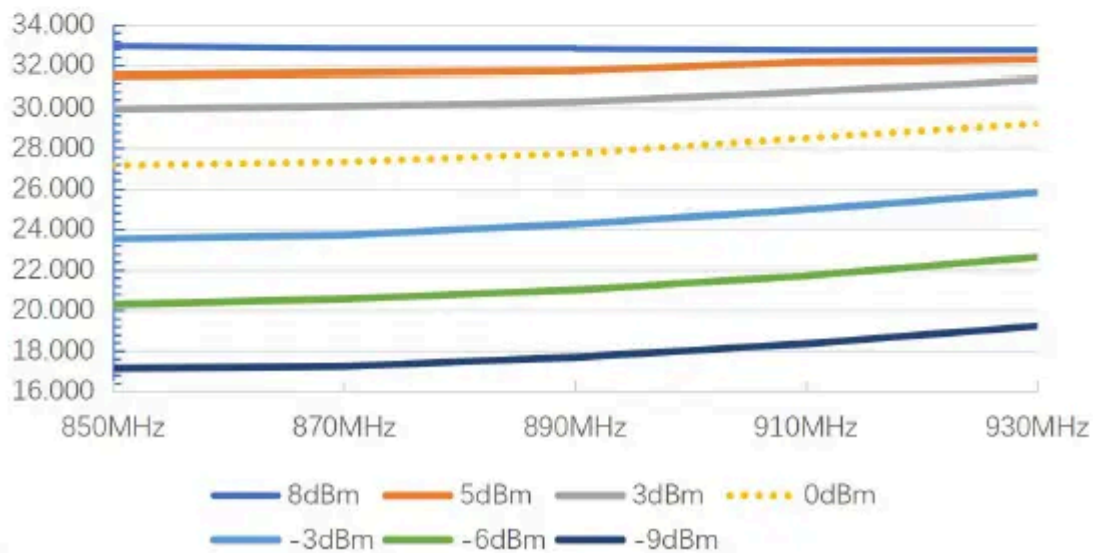
weight| 5.1g| ± 0.1 g

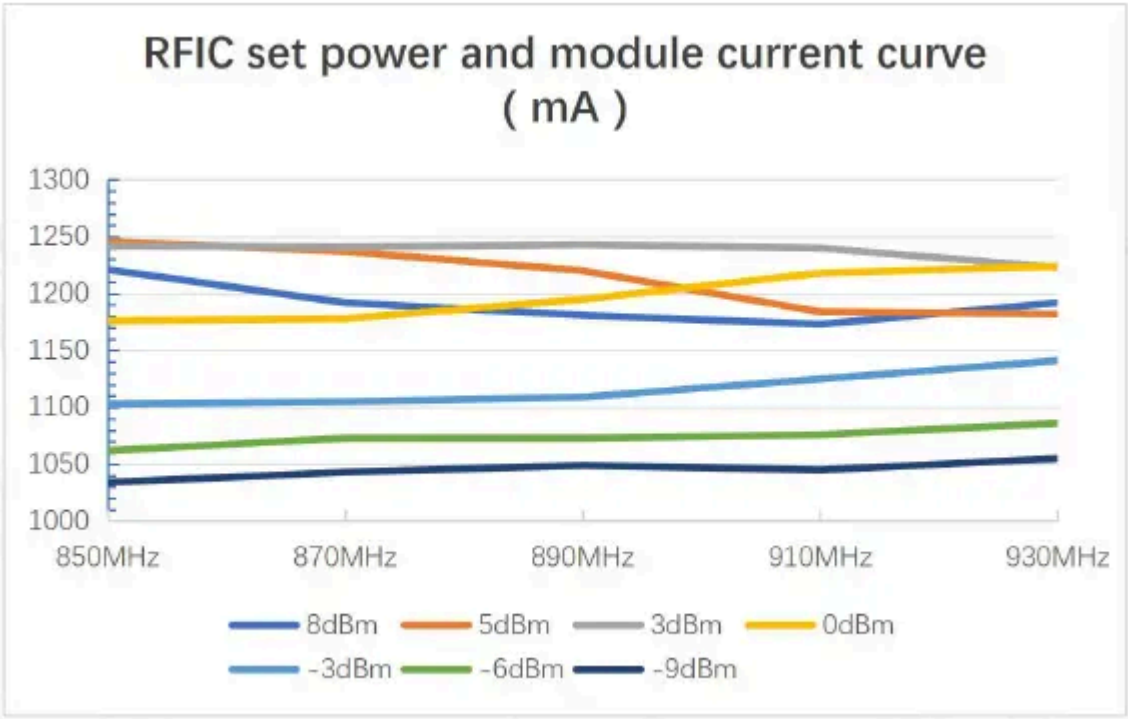
TX Power Curve

RFIC set power and module actual output power curve (dBm)

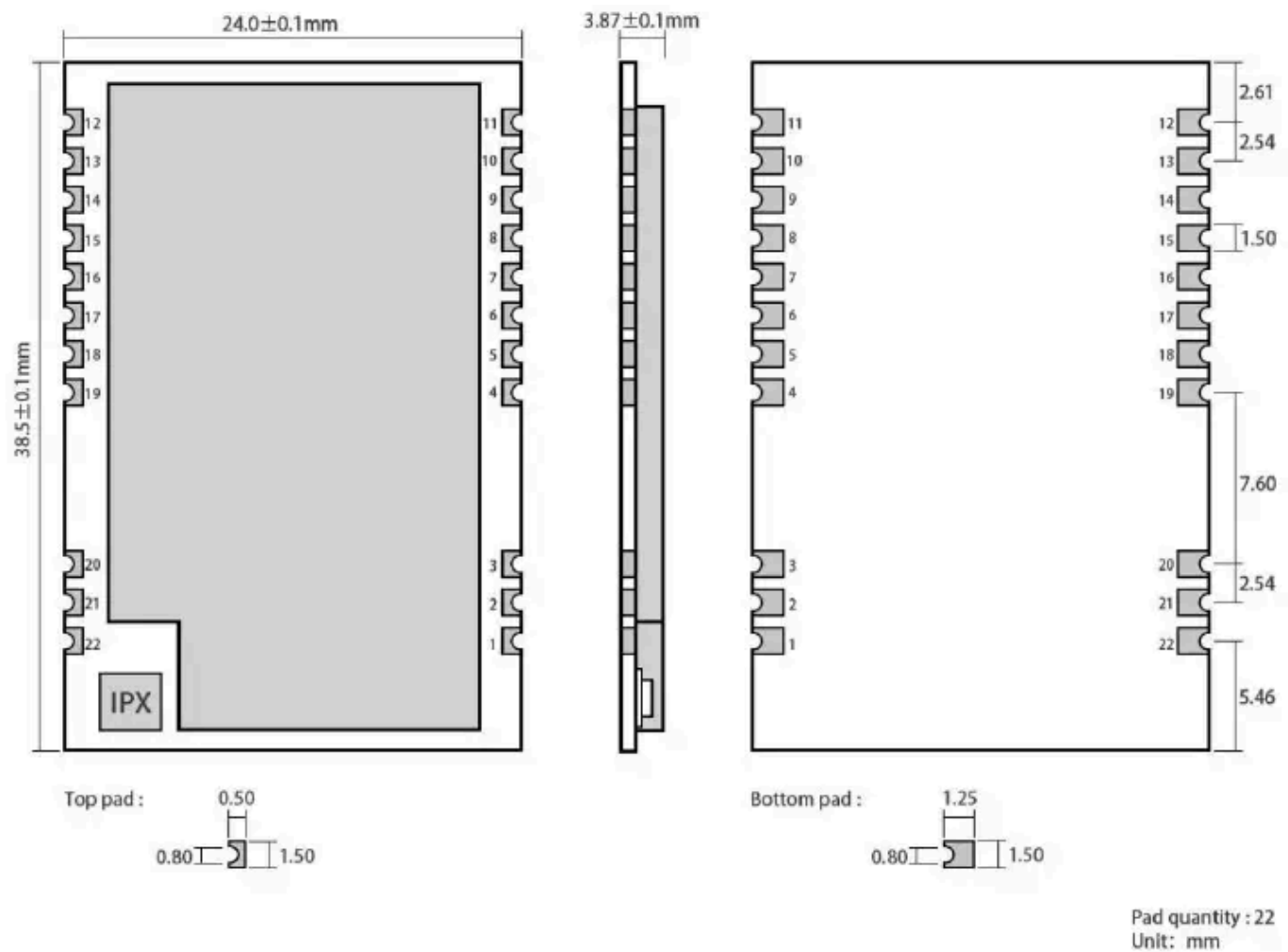


RFIC set power and module actual output power curve (dBm)





Mechanical Dimensions and Pin Definitions



Pin	Pin name	Pin direction	Pin usage
1	GND		Ground wire, connected to the power reference ground

Pin	Pin name	Pin direction	Pin usage
2	GND		Ground wire, connected to the power reference ground
3	GND		Ground wire, connected to the power reference ground
4	GND		Ground wire, connected to the power reference ground
5	GND		Ground wire, connected to the power reference ground

6

| ****

RXEN

| ****

enter

| The radio frequency switch receives the control pin, connects to the external microcontroller IO, and

is active at high level.

7| TXEN| enter| RF switch emission control pin, connected to external microcontroller IO or DIO2, active at high level

8| DIO2| input Output| Configurable general IO port (see SX1262 manual for details)

9| VCC| | Power supply, range 3.3 ~ 5.5V (it is recommended to add external ceramic filter capacitors)

10| VCC| | Power supply, range 3.3 ~ 5.5V (it is recommended to add external ceramic filter capacitors)

11| GND| | Ground wire, connected to the power reference ground

12| GND| | Ground wire, connected to the power reference ground

13| DIO1| input Output| Configurable general IO port (see SX1262 manual for details)

14| BUSY| output| Used for status indication (see SX1262 manual for details)

15| NRST| enter| Chip reset trigger input pin, active low level

16| MISO| output| SPI data output pin

17| MOSI| enter| SPI data input pin

18| SCK| enter| SPI clock input pin

19| NSS| enter| Module chip select pin, used to start an SPI communication

20| GND| | Ground wire, connected to the power reference ground

21| ANT| | Antenna interface, stamp hole (50 Ω characteristic impedance)

22| GND| | Ground wire, connected to the power reference ground

Notes For Hardware And Software

Hardware design

- It is recommended to use a DC regulated power supply to power the module. The power supply ripple coefficient should be as small as possible, and the module must be reliably grounded;
- Please pay attention to the correct connection of the positive and negative poles of the power supply. Reverse connection may cause permanent damage to the module;
- Please check the power supply to ensure that it is within the recommended power supply voltage. If it exceeds the maximum value, it will cause permanent damage to the module;
- Please check the stability of the power supply. The voltage cannot fluctuate greatly and frequently;
- When designing the power supply circuit for the module, it is often recommended to reserve more than 30% margin, so that the whole machine can work stably for a long time;
- The module should be kept as far away as possible from power supplies, transformers, high-frequency wiring and other parts with high electromagnetic interference;
- High-frequency digital traces, high-frequency analog traces, and power traces must be avoided under the module. If it is absolutely necessary to pass under the module, assuming that the module is welded on the Top Layer, lay copper ground on the Top Layer of the module contact part (all copper and well grounded), it must be close to the digital part of the module and routed on the Bottom Layer ;
- Assuming that the module is welded or placed on the Top Layer, it is also wrong to route traces randomly on the Bottom Layer or other layers, which will affect the module's spurious and receiving sensitivity to varying degrees ;
- Assuming that there are devices with large electromagnetic interference around the module, which will also greatly affect the performance of the module, it is recommended to stay away from the module according to the intensity of the interference. If the situation allows, appropriate isolation and shielding can be done;
- Assuming that there are traces with large electromagnetic interference around the module (high-frequency digital, high-frequency analog, power traces), it will also greatly affect the

performance of the module. It is recommended to stay away from the module according to the intensity of the interference. If the situation allows, you can make appropriate adjustments. isolation and shielding;

- If the communication line uses 5V level, a 1k-5.1k resistor must be connected in series (not recommended, there is still a risk of damage) ;
- Try to stay away from some TTL protocols whose physical layer is also 2.4GHz, such as USB3.0;
- The antenna installation structure has a great impact on module performance. Make sure the antenna is exposed, preferably vertically upward. When the module is installed inside the case, you can use a high-quality antenna extension cable to extend the antenna to the outside of the case;
- The antenna must not be installed inside a metal shell, as this will greatly reduce the transmission distance.
- to add a 200R protection resistor to the RXD/TXD of the external MCU.

Software writing

```
void Ebyte_E22x_Init( void )
{
    /* IO复位+CS唤醒 模块*/
    Ebyte_E22x_Reset( );
    Ebyte_E22x_Wakeup( );

    /* 状态机设定 */
    OperatingMode = MODE_STDBY_RC;

    /* 进入 STDBY_RC 待机配置模式 */
    Ebyte_E22x_SetStandby( STDBY_RC );

    /* 启用TCXO(由DIO3控制) 并尝试进入STDBY_XOSC模式 3.3V输出 PLL会自动补偿33.4pf电容器 */
    Ebyte_E22x_SetDio3AsTcxoCtrl ( TCXO_CTRL_3_3V, 320 );
    Ebyte_E22x_WriteRegister ( REG_XTA_TRIM, 0x2F );

    /* 重新计算修正时钟 */
    CalibrationParams_t calibrationParams;
    calibrationParams.Value = 0xFF;
    Ebyte_E22x_Calibrate( calibrationParams );

    /* 进入 STDBY_XOSC 待机配置模式 */
    Ebyte_E22x_SetStandby ( STDBY_XOSC );

    /* 是否启用 DIO2 自动控制TXEN */
    Ebyte_E22x_SetDio2AsRfSwitchCtrl( 1 );

    /* 选择内部电压调节器模式 高效DC-DC */
    Ebyte_E22x_SetRegulatorMode( USE_DCDC );

    /* 内部FIFO读写地址复位 0x00 */
    Ebyte_E22x_SetBufferBaseAddress( 0x00, 0x00 );
}
```

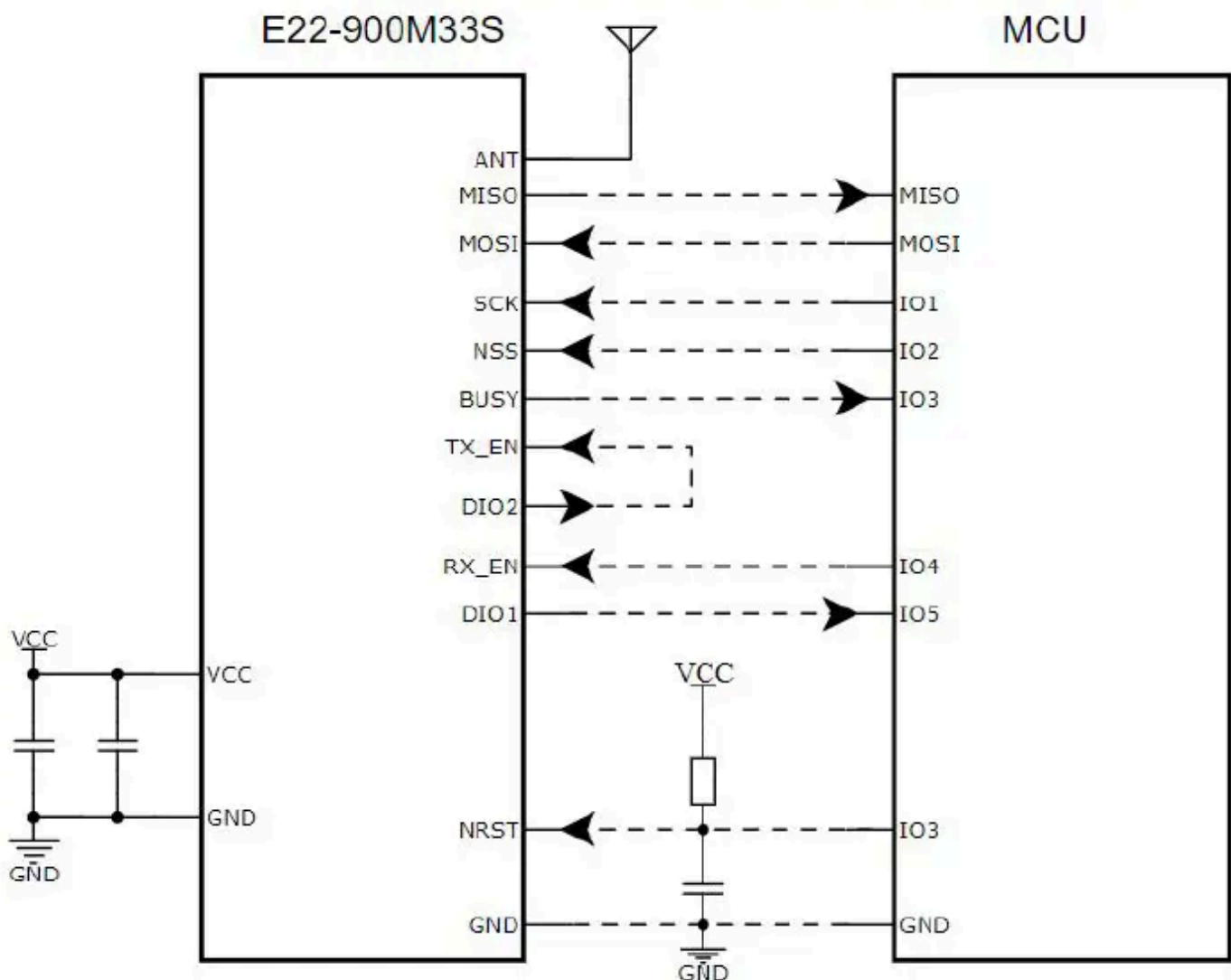
- This module is SX1268/SX1262+PA+LNA , and its driving method is completely the same as SX1268/SX1262 . Users can operate it according to the SX1268/SX1262 chip

manual;It should be noted that the TX_EN of PA_EN and RF_switch are connected together, and the RX_EN of LNA_EN and RF_switch are connected together, so users only need to control TX_EN when sending, and use TX_EN when receiving.

- DIO1 and DIO2 are general-purpose IO ports that can be configured into a variety of functions; DIO2 can be connected to TXEN, not connected to the IO port of the MCU, and is used to control radio frequency switch emission. For details, see the SX1262 manual. If not used , it can be left floating;RX_EN and TX_EN cannot be high at the same time. The level states must be opposite.
- is used internally to power the 32MHz TCXO crystal oscillator (DIO3 is configured to output 1.8V) .

MCU Connection

Recommended connection



Frequently Asked Questions

Transmission distance is not ideal

- When there are straight-line communication obstacles, the communication distance will be correspondingly attenuated ;
- Temperature, humidity, and co-channel interference will cause the communication packet loss rate to increase ;
- The ground absorbs and reflects radio waves, and the test effect is poor when close to the ground ;
- Seawater has a strong ability to absorb radio waves, so the seaside test results are poor ;
- If there are metal objects near the antenna, or if it is placed in a metal case, the signal attenuation will be very serious ;
- The power register setting is wrong and the air rate is set too high (the higher the air rate, the closer the distance) ;
- The low voltage of the power supply at room temperature is lower than the recommended value. The lower the voltage, the smaller the power generated ;
- There is a poor match between the antenna and the module or there is a problem with the quality of the antenna itself.

Modules are easily damaged

- Please check the power supply to ensure that it is within the recommended power supply voltage. If it exceeds the maximum value, it will cause permanent damage to the module ;
- Please check the stability of the power supply. The voltage cannot fluctuate greatly and frequently ;
- Please ensure anti-static operation during installation and use, as high-frequency devices are sensitive to static electricity ;
- Please ensure that the humidity during installation and use should not be too high, as some components are humidity-sensitive devices ;
- If there are no special needs, it is not recommended to use it at too high or too low temperature.

Bit error rate too high

- If there is co-channel signal interference nearby, stay away from the interference source or modify the frequency or channel to avoid interference;
- The clock waveform on SPI is not standard . Check whether there is interference on the SPI line . The SPI bus line should not be too long;

- An unsatisfactory power supply may also cause garbled code, so be sure to ensure the reliability of the power supply;
- Poor quality or too long extension cords and feeders can also cause a high bit error rate.

Welding Operation Guidance

Reflow soldering temperature

Reflow soldering curve characteristics| Lead process assembly| Lead-free process assembly

---|---|---

Preheat/keep warm

| Minimum temperature (Tsmin)| 100°C| 150°C

Maximum temperature (Tsmax)| 150°C| 200°C

Time (Tsmin~Tsmin)

|

60-120 seconds

|

60-120 seconds

Temperature rise slope (TL~Tp)| 3°C/second, maximum| 3°C/second, maximum

Liquidus temperature (TL)| 183°C| 217°C

Holding time above TL| 60~ 90 seconds| 60~ 90 seconds

Package peak temperature Tp

| Users should not exceed the

temperature indicated on the

| Users should not exceed the

temperature indicated on the

| product’s “Moisture Sensitivity” label.| product’s “Moisture Sensitivity” label.

---|---|---

The time (Tp) within 5°C of the specified classification temperature (Tc), see the figure below|

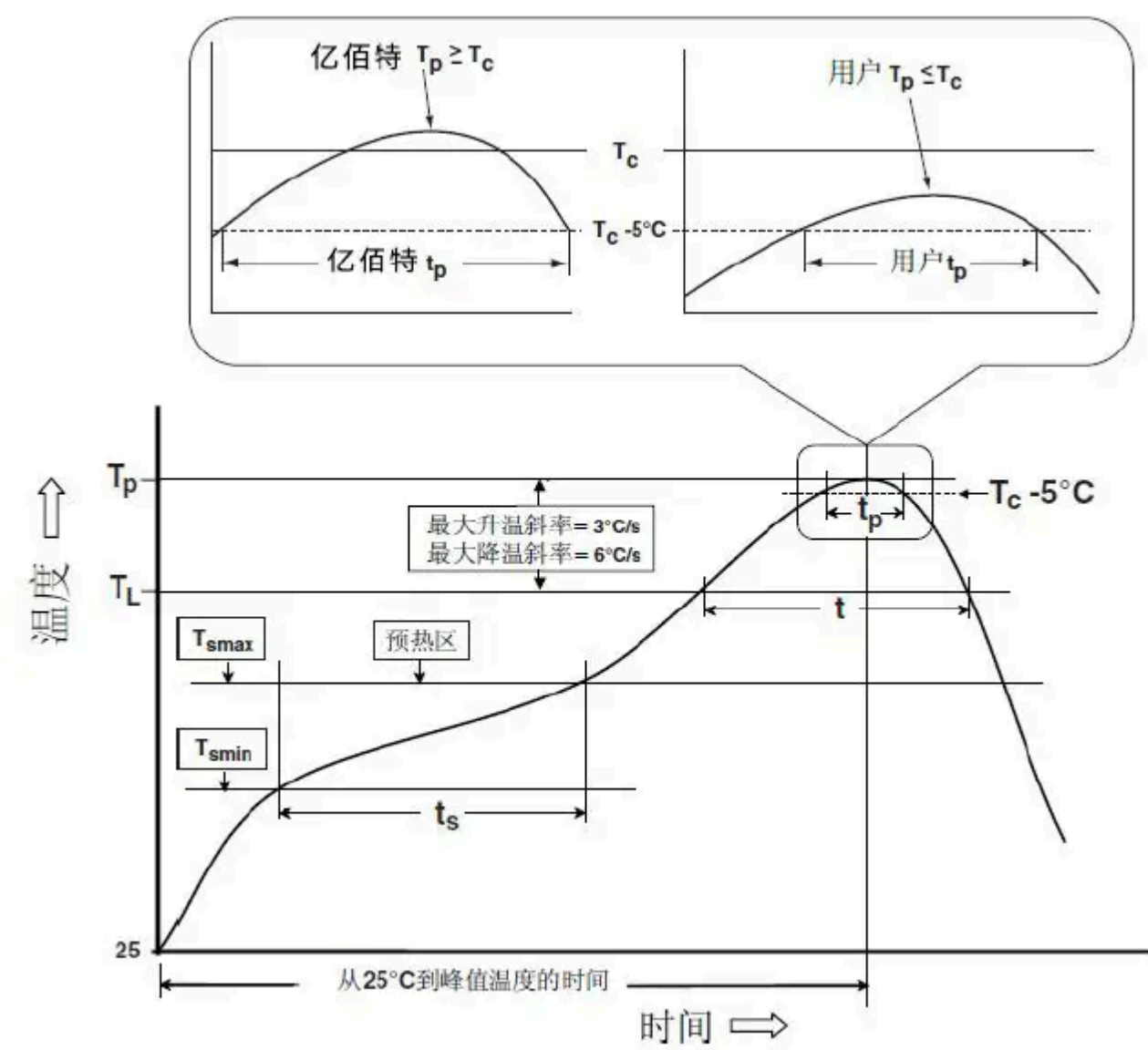
20 seconds| 30 seconds

Cooling slope (Tp~TL)| 6°C/second, maximum| 6°C/second, maximum

Time from room temperature to peak temperature| 6 minutes, maximum| 8 minutes, maximum

※The peak temperature (Tp) tolerance definition of the temperature curve is the upper limit of the user

Reflow soldering curve



Related Models

Product number

| ****

Chip solution

| carrier frequency

Hz

| Transmit power

dBm

| ****

Test distance km

| ****

Package form

| ****

Product Size mm

| ****

Communication Interface

---|---|---|---|---|---|---

[E22-400M22S](#)| SX1262| 433/470M| twenty two| 7| patch| 1420| *SPI* _

[E22-900M22S](#)| SX1262| 868/915M| *twenty two*| 7| *patch*| 1420| SPI

[E22-900M33S](#)| SX1262| 433/470M| 3 3| 1 6| *patch*| 24*38.5| *SPI*

[E22-900M30S](#)| SX1262| 868/915M| 30| 12| patch| 2438.5| *SPI* _

[E22-230T22S](#)| SX1262| 230M| *twenty two*| 5| *patch*| 1626| TTL

[E22-400T22S](#)| SX1262| 433/470M| twenty two| 5| patch| 1626| *TTL*

---|---|---|---|---|---|---

[E22-900T22S](#)| SX1262| 868/915M| *twenty two*| 5| *patch*| 1626| TTL

[E22-230T30S](#)| SX1262| 230M| 30| 1 0| patch| 2540.5| *TTL*

[E22-400T30S](#)| SX1262| 433/470M| 30| 1 0| *patch*| 2540.5| TTL

[E22-900T30S](#)| SX1262| 868/915M| 30| 1 0| patch| 25*40.5| TTL

Antenna Guide

Antenna recommendations

Antennas play an important role in the communication process. Often poor- quality antennas will have a great impact on the communication system. Therefore, our company recommends some antennas as antennas that support our wireless modules and have excellent performance and reasonable prices.

Product number| type| frequency band Hz| interface| Gain dBi| high m m| feeder cm|

Features

---|---|---|---|---|---|---

TX433-NP-4310

|
flexible antenna
| 433M|
welding
|
2.0 _
|
43.8*9.5
|
—
| Built-in flexible, FPC soft antenna

TX433-JZ-5

|
glue stick antenna
|
433M
|
SMA-J
|
2.0 _
|
52
|

—

| Ultra-short straight, omnidirectional antenna

TX433-JZG-6

|

glue stick antenna

|

433M

|

SMA-J

|

2.5

|

6 2

|

—

| Ultra-short straight, omnidirectional antenna

TX433-JW-5

|

glue stick antenna

|

433M

|

SMA-J

|

2.0 _

|

50

|

—

| Bend glue stick, omnidirectional antenna

[TX433-JWG-7](#)

|

glue stick antenna

|

433M

|

SMA-J

|

2.5

|

75

|

—

| Bend glue stick, omnidirectional antenna

[TX433-JK-11](#)

|

glue stick antenna

|

433M

|

SMA-J

|

2.5

|

110

|

—

| Bendable glue stick, omnidirectional antenna

[TX433-JK-20](#)

|

glue stick antenna

|

433M

|

SMA-J

|

3.0 _

|

210

|

—

| Bendable glue stick, omnidirectional antenna

[TX433-XPL – 100](#)

| suction cup antenna|

433M

|

SMA-J

|

3.5

|

18 5

|

100

| Small suction cup antenna, cost-effective

[TX433-XP-200](#)

| suction cup antenna|

433M

|

SMA-J

|

4.0 _

|

19 0

|

200

| Neutral suction cup antenna, low loss

[TX433-XP-300](#)

| suction cup antenna|

433M

|

SMA-J

|

6.0 _

|

96 5

|

300

| Large suction cup antenna, high gain

[TX490-JZ-5](#)

|

glue stick antenna

| 4 7 0 /490

M

|

SMA-J

|

2.0

|

50

|

—

| Ultra-short straight, omnidirectional antenna

[TX490-XPL – 100](#)

| suction cup antenna| 4 7 0 /

490M

|

SMA-J

|

3.5

|

12 0

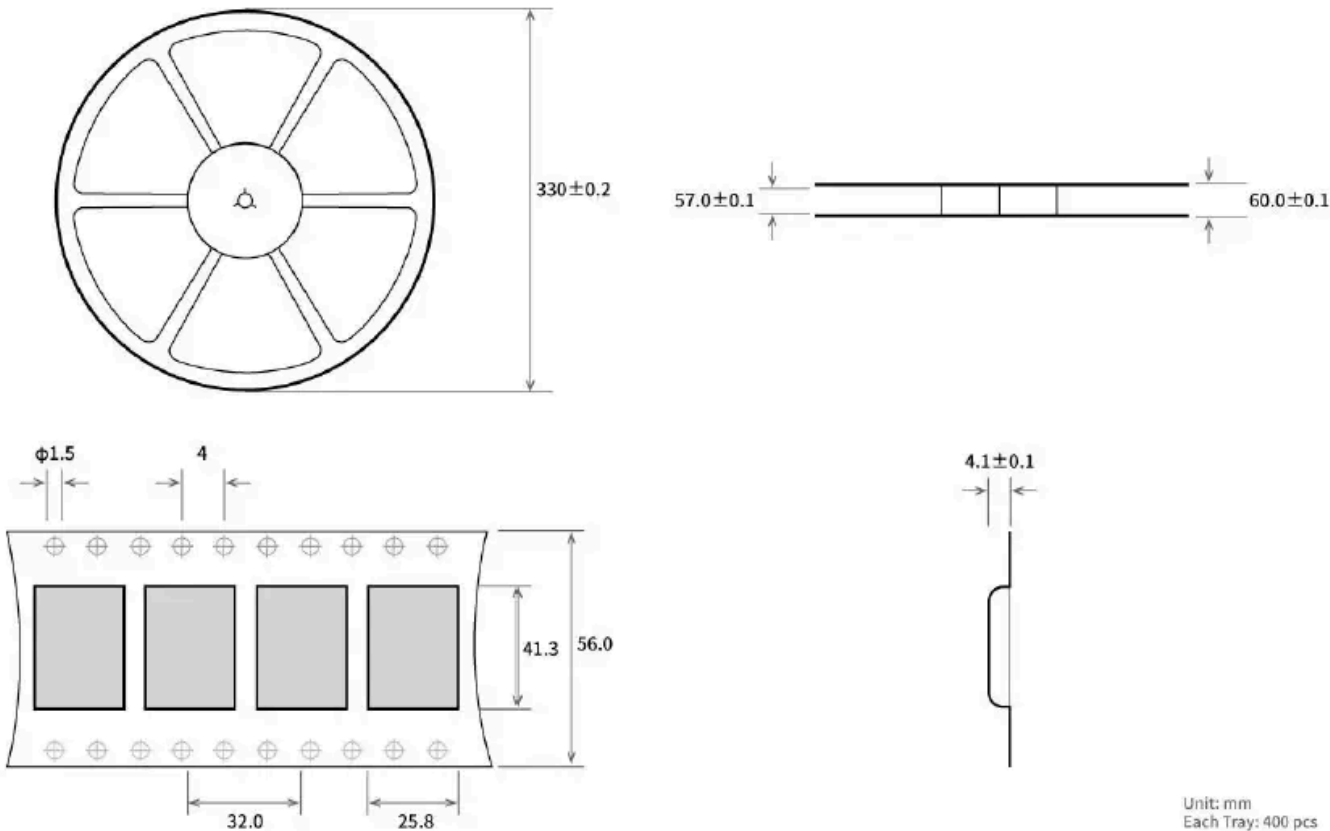
|

100

| Small suction cup antenna, cost-effective

---|---|---|---|---|---|---

Standard Package



Revise History

Version	Revision date	Revision Notes	Maintenance man
1.0	2024-1-24	Manual release	Hao

Contact Us

- Technical support: support@cdebyte.com
- Documents and RF Setting download link: <https://www.ru-ebyte.com>
- Thank you for using Ebyte products! Please contact us with any questions or suggestions: info@cdebyte.com

- **Address:** B5 Mould Industrial Park, 199# Xiqu Ave, High tech Zone, Chengdu, Sichuan, China

Chengdu Ebyte Electronic Technology Co.,Ltd.

References

- [Китайские производители беспроводных модемов Lora, поставщики промышленных терминалов IoT](#)

Read User Manual Online (PDF format)



 Loading.....

Download This Manual (PDF format)

[Download this manual >>](#)

ebyte User Manuals

EBYTE E70-433NW30S 433MHz 1W St... 📁 ebyte	EBYTE E610-433T30S 433MHz 1W Wir... 📁 ebyte
EBYTE E32-900T20D SX1276 868MHz/... 📁 ebyte	EBYTE E22-900M33S SX1262 868/915 ... 📁 ebyte
EBYTE E52-400/900NW22S LoRa MES... 📁 ebyte	EBYTE E73-2G4M04S1F BLE Module Us... 📁 ebyte

EBYTE AM11-12W05V 12W Low Power ...

📁 ebyte

EBYTE E01C-ML01S Si24R1 2.4GHz SPI ...

📁 ebyte

EBYTE E95-DTU 400SL22P-485 Wireles...

📁 ebyte

EBYTE E32-900M30S 915MHz SPI Wire...

📁 ebyte

gSI4463 915MHz 1W SMD Wireless Mo...

📁 ebyte

EBYTE E78-400M22S1C LPWAN Low P...

📁 ebyte

EBYTE E842-DTU Wireless Modem Use...

📁 ebyte

EBYTE E19-868M20S 100mW SMD Wir...

📁 ebyte

EBYTE E73-2G4M04S1B nRF52832 BLE ...

📁 ebyte

EBYTE E01C-ML01SP4 Wireless Modul...

📁 ebyte

EBYTE E49-400M30S Wireless Module...

📁 ebyte

EBYTE DM21-36W12S 36W Step-Down ...

📁 ebyte

EBYTE E32-400M30S SX1278 433/470...

📁 ebyte

EBYTE E07-900M10S CC1101 904MHz-...

📁 ebyte

Related Manuals

NEXTIME 5190 Black Digital Clock Instr...

📁 NEXTIME

Ematic All-in-One Easy Wear SMART ...

📁 ematic

LG UltraGear 27GN800P-B Instruction ...

📁 LG

iSetchi 3-in-1 Dual Coils Wireless Chargi...

📁 iSetchi

WAGNER WiiM Amp Set Up Instructions

📁 Wagner

Goodman CK36-1A Room Air Condition...

📁 Goodman

REGIN RC-C3DFOC Pre Programmed ...

📁 REGIN

ASCOT K1 Stainless Steel Electric Tea K...

📁 ASCOT

Intel Interlaken 2nd Generation Agilex 7 ...

📁 Intel

FISHER and PAYKEL CPV3-486GD-L 48...

📁 FISHER and PAYKEL

Kodak PIXPRO SL25 Smart Lens Camer...

📁 Kodak

Drager PSS 3000 Series Self Contained ...

📁 Drager

VEGETA KZW-U10217B 10 Ports Powere...

📁 VEGETA

CHARGE AMPS Dawn EV Charger Use...

📁 CHARGE AMPS

ZEBRONICS ZEB Dj Speaker User Manual

📁 ZEBRONICS

Verbatim External Slimline Blu-Ray Writ...

📁 Verbatim

Honeywell Home HS9HUBGPRS Wirele...

📁 Honeywell Home

DENVER ELECTRONICS CR-419 MK2 P...

📁 Denver Electronics

SANYO AHX0952 Series Split System A...

📁 SANYO

FEIT Electric S9DFL Smart Dual Head ...

📁 Feit Electric

VOLLRATH 38711 Cafeteria Breath Guar...

📁 VOLLRATH

ENYA MUSIC Nova Go Sonic 38 Inch F...

📁 ENYA MUSIC

HYUNDAI CY 300MAX Smart Watch In...

📁 HYUNDAI

Travel Buggy DASH Ultra Lite Power Ch...

📁 Travel Buggy

RANCEO PF3 Flashlight User Manual

📁 RANCEO

BEGA 31 053 Wall Luminaire Instruction...

📁 BEGA

BOMANN KSG 7288 Household Bevera...

📁 BOMANN

Bewinner1 OR1142-UNI-BIG Combo Uni...

📁 Bewinner1

[AMVEX VR-C Vacuum Regulator Instru...](#)

📁 AMVEX

[TURNART B08YYY1LDF Wheel Clamp ...](#)

📁 TURNART

[Contact Us](#) [Privacy Policy](#) 7.790ms